

SmartVAPT Security Assessment Report

ICT Academy of Kerala - Cyber Security Analyst (VAPT)

Vulnerability Assessment Report

Target: <https://example.com>

Date: September 29, 2026

1. Executive Summary

AI unavailable: GEMINI_API_KEY not configured. Set env var or edit config.py

Overall Risk: **Unknown (0/10)**

2. Top Findings

No findings reported.

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3. Penetration-Test Details & Exploitation Options

3.1 Test Details & Methodology

Target: https://example.com Scan ID: 20260929_032224 Date: 2026-09-29 03:22

Ports scanned: Nmap Top 1000 ports (selection: top1000)

Network: nmap with version detection (-sV), default scripts (-sC) and the vulners NSE script; HTTP services were located via -sV and directory brute-forcing (gobuster dir) was executed against the detected HTTP port(s) rather than assuming port 80.

Web audit base URL (nmap-confirmed): http://example.com

No HTTP-speaking ports were detected by nmap -sV.

Nuclei (default template scanner): unavailable this run — nuclei binary not found. Install it: sudo apt install nuclei (or download from https://github.com/projectdiscovery/nuclei). Install the 'nuclei' binary to enable it.

Rules of engagement: all steps above are non-destructive checks performed within the authorized scope of this assessment.

3.2 Exploitation Options (authorized verification only)

1. [HIGH] No HTTP -> HTTPS redirect

Attack paths : Downgrade to weak cipher (SWEET32/BEAST class issues); Interception when certificate is invalid/expired

Difficulty : Moderate

Verify : openssl s_client -connect <host>:<port> -tls1_1 # should FAIL if hardened

Verify : nmap --script ssl-enum-ciphers -p <port> <host>

Tools : openssl s_client, testssl.sh, sslyze

2. [MEDIUM] Missing security header: Strict-Transport-Security

Attack paths : Clickjacking (missing X-Frame-Options/CSP frame-ancestors); MIME-confusion scripting (missing X-Content-Type-Options); SSL-stripping (missing HSTS)

Difficulty : Informational

Verify : curl -skI <url> | grep -iE 'strict-transport|content-security|x-frame|x-content-type'

Verify : Open the page inside an <iframe> on a test page to confirm framing is allowed.

Tools : curl, browser dev-tools, nuclei (-t http/headers/)

3. [MEDIUM] Missing security header: Content-Security-Policy

Attack paths : Clickjacking (missing X-Frame-Options/CSP frame-ancestors); MIME-confusion scripting (missing X-Content-Type-Options); SSL-stripping (missing HSTS)

Difficulty : Informational

Verify : curl -skI <url> | grep -iE 'strict-transport|content-security|x-frame|x-content-type'

Verify : Open the page inside an <iframe> on a test page to confirm framing is allowed.

Tools : curl, browser dev-tools, nuclei (-t http/headers/)

4. [MEDIUM] Missing security header: X-Frame-Options

Attack paths : Clickjacking (missing X-Frame-Options/CSP frame-ancestors); MIME-confusion scripting (missing X-Content-Type-Options); SSL-stripping (missing HSTS)

Difficulty : Informational

Verify : curl -skI <url> | grep -iE 'strict-transport|content-security|x-frame|x-content-type'

Verify : Open the page inside an <iframe> on a test page to confirm framing is allowed.

Tools : curl, browser dev-tools, nuclei (-t http/headers/)

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5. [MEDIUM] Missing security header: X-Content-Type-Options

Attack paths : Clickjacking (missing X-Frame-Options/CSP frame-ancestors); MIME-confusion scripting (missing X-Content-Type-Options); SSL-stripping (missing HSTS)

Difficulty : Informational

Verify : `curl -skI <url> | grep -iE 'strict-transport|content-security|x-frame|x-content-type'`

Verify : Open the page inside an <iframe> on a test page to confirm framing is allowed.

Tools : curl, browser dev-tools, nuclei (-t http/headers/)

6. [MEDIUM] Missing security header: Referrer-Policy

Attack paths : Clickjacking (missing X-Frame-Options/CSP frame-ancestors); MIME-confusion scripting (missing X-Content-Type-Options); SSL-stripping (missing HSTS)

Difficulty : Informational

Verify : `curl -skI <url> | grep -iE 'strict-transport|content-security|x-frame|x-content-type'`

Verify : Open the page inside an <iframe> on a test page to confirm framing is allowed.

Tools : curl, browser dev-tools, nuclei (-t http/headers/)

7. [MEDIUM] Missing security header: Permissions-Policy

Attack paths : Clickjacking (missing X-Frame-Options/CSP frame-ancestors); MIME-confusion scripting (missing X-Content-Type-Options); SSL-stripping (missing HSTS)

Difficulty : Informational

Verify : `curl -skI <url> | grep -iE 'strict-transport|content-security|x-frame|x-content-type'`

Verify : Open the page inside an <iframe> on a test page to confirm framing is allowed.

Tools : curl, browser dev-tools, nuclei (-t http/headers/)

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4. Technical Remediation

No technical remediation steps provided.

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5. Appendix: Scan Details

A. Target Information

Target: https://example.com
Scan ID: 20260929_032224

B. Network Scan

Error: [Errno 2] No such file or directory: 'nmap'

C. Web Application Scan

Server: cloudflare

Missing security headers (6):

- Strict-Transport-Security
- Content-Security-Policy
- X-Frame-Options
- X-Content-Type-Options
- Referrer-Policy
- Permissions-Policy

Present security headers (0):

Detected technologies (1):

- cloudflare (via header:Server)

Redirect chain (1 hops):

200 http://example.com/

D. DNS / Email Security

Domain: example.com

SPF Present: Yes

Record: v=spf1 -all

DMARC Present: Yes

Record: v=DMARC1;p=reject;sp=reject;adkim=s;aspf=s

Policy: reject

I. Full Findings List

1. [HIGH] No HTTP -> HTTPS redirect

Target: http://example.com

Evidence: Final URL: http://example.com/

Remediation: Disable TLS 1.0/1.1; require TLS 1.2+ with AEAD cipher suites.; Renew certificates expiring within 30 days and automate renewal (ACME).; Enable HSTS after HTTPS is fully deployed.

2. [MEDIUM] Missing security header: Strict-Transport-Security

Target: http://example.com

Evidence: GET http://example.com -> no 'Strict-Transport-Security' header

Remediation: Identify the web server / reverse proxy in front of the application.; Add the missing security header globally so every response carries it.; Verify with a re-scan that the header is present on all endpoints.

3. [MEDIUM] Missing security header: Content-Security-Policy

Target: http://example.com

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Evidence: GET http://example.com -> no 'Content-Security-Policy' header

Remediation: Identify the web server / reverse proxy in front of the application.; Add the missing security header globally so every response carries it.; Verify with a re-scan that the header is present on all endpoints.

4. [MEDIUM] Missing security header: X-Frame-Options

Target: http://example.com

Evidence: GET http://example.com -> no 'X-Frame-Options' header

Remediation: Identify the web server / reverse proxy in front of the application.; Add the missing security header globally so every response carries it.; Verify with a re-scan that the header is present on all endpoints.

5. [MEDIUM] Missing security header: X-Content-Type-Options

Target: http://example.com

Evidence: GET http://example.com -> no 'X-Content-Type-Options' header

Remediation: Identify the web server / reverse proxy in front of the application.; Add the missing security header globally so every response carries it.; Verify with a re-scan that the header is present on all endpoints.

6. [MEDIUM] Missing security header: Referrer-Policy

Target: http://example.com

Evidence: GET http://example.com -> no 'Referrer-Policy' header

Remediation: Identify the web server / reverse proxy in front of the application.; Add the missing security header globally so every response carries it.; Verify with a re-scan that the header is present on all endpoints.

7. [MEDIUM] Missing security header: Permissions-Policy

Target: http://example.com

Evidence: GET http://example.com -> no 'Permissions-Policy' header

Remediation: Identify the web server / reverse proxy in front of the application.; Add the missing security header globally so every response carries it.; Verify with a re-scan that the header is present on all endpoints.